HW1 3. Soru Çözümünde Düzeltme

$(k=1;\ k \leq j;\ k=k*3) \rightarrow \log_3 j$

kere çalışır.

$$= \sum_{i=1}^{2n} \sum_{j=1}^{i} \log_3 j$$

$$= \sum_{i=1}^{2n} \log_3 1 + \log_3 2 + \dots + \log_3 i$$

$$= \sum_{i=1}^{2n} \left(\log_3 i!\right)$$

⋮

HW #2 Solutions

1)

a) We will solve this like solving a differential equation.

$r^2 - 9r + 20 = 0 \qquad T(0) = 0,\ T(1) = 1$

$r^2 - 9r + 20 = (r-5)(r-4)$ so roots are 4 and 5.

Hence $T(n) = c_1 5^n + c_2 4^n$

$T(0) = c_1 + c_2 = 0 \implies c_2 = -c_1$

$T(1) = 5c_1 + 4c_2 = 1 = 5c_1 - 4c_1 = c_1$

So $c_1 = 1$ and $c_2 = -1$.

$T(n) = 5^n - 4^n$ so $\Theta(5^n)$.

aslında burada göstermemiz gerekiyor.

b) $T(n) = 2T(n/2) + n\log n, \quad T(1) = 1$

$n = 2^k$ olsun.

$$T(2^k) = 2T(2^{k-1}) + \frac{2^k}{k}$$

$$= 2^2 \, T(2^{k-2}) + 2^k \left(\frac{1}{k} + \frac{1}{k-1} \right)$$

$$\vdots$$

$$= 2^i \cdot T(2^{k-i}) + 2^k \left(\frac{1}{k} + \frac{1}{k-1} + \dots + \frac{1}{k-i+1} \right)$$

$i == k$ olduğu zaman

$$T(2^k) = 2^k \, T(1) + 2^k \left(\sum_{a=1}^{k} \frac{1}{a} \right)$$

$$= n + n \left(\sum_{a=1}^{k} \frac{1}{a} \right)$$

$H_k = \sum_{a=1}^{k} \frac{1}{a}$ is called harmonic number.

and $H_k = \Theta(\log k)$ (you can find proof on internet)

so

$T(n) = n + n H_{\log n} = \Theta(n \log \log n)$.

1-c)

$$T(n) = T(n-1) + n \quad , \quad T(1) = 1$$

$$= T(n-2) + (n-1) + n$$

$$\vdots$$

$$= T(1) + 2 + 3 + 4 + \ldots + n$$

bunun doğruluğunu induction ile gösterebiliriz.

$$= \frac{n.(n+1)}{2} = \Theta(n^2)$$

2)

$$T(i,j) = \begin{cases} 1 & \text{if } (i < 1) \text{ or } (j < 1) \\ T((i-1),(j-1)) & \text{if } A[i] == B[j] \\ T(i, j-1) + T(i-1, j) & \text{else} \end{cases}$$

worst case is the last case. So if always the last case it will be the worst case. Let's draw tree. Depth will be $(i+j)$.

So complexity is $\boxed{\Theta(2^{i+j-1})}$

$T(i,j)$

$T(i,j-1)$ $T(i-1,j)$

$T(i,j-2)$ $T(i-1,j-1)$ $T(i-2,j)$ $T(i-1,j-1)$

3) You can find the answer on the internet.

Search for "shifted sorted array logn"

4)

Similar to quicksort partition the array according to 3's and not 3 first.

3	2	2	1	3	1	2	3	1	2

→ go till find 3

← go till find something not 3

swap the 3 and not 3 number.

so we have

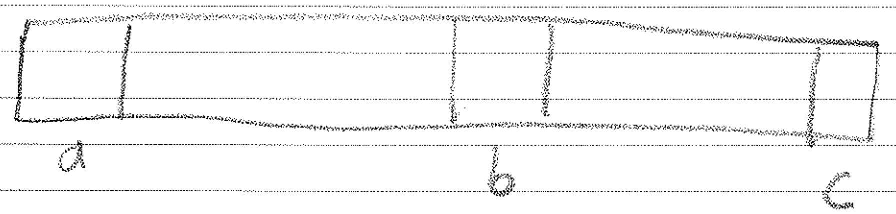
1	2	1	2	1	1	3	3	3

→ go till find 2

← go till find 1

now do the same thing for 1 and 2 here.

5) This is somewhat similar to question 3.

A 

if $b - a == A[b] - A[a]$:

search at the other part.

else :

pick a median element at left part and do the same recursively.